

M. J. Biopharm Private Limited Plot No. 18, International Biotech Park, Hinjawadi, Phase II, Pune 411057.	
Format for Attachment	

Title: Outcome of Recommendation action -I of deviation occurred during calibration of TOC Analyzer (Instrument ID: I/QC/018)	
Attachment No: <u>XII</u>	Ref. document No: DEV/QC/25/010

Outcome of Recommended Action I:

As per recommended action – I, performed the re-dilution of 1,4-Benzoquinone stock standard solution (50000 ppb) as below;

Re-Dilution: Taken 1 ml of 1,4 - Benzoquinone stock standard solution (50000 ppb) and diluted to 100 ml with purified water. (1,4 - Benzoquinone Working Standard (500 ppb).

Further, filled the Re-diluted standard solution and existing standard solution of 1,4 - Benzoquinone standard solutions (500 ppb) in two individual 40 ml cleaned, dried TOC vial and kept vial in TOC autosampler tray.

As per the recommended action I, analyst injected 1,4 - Benzoquinone standard solutions (500 ppb) as per below sequence in the TOC analyzer to evaluate the potential dilution error and to ensure the proper functioning of instrument. The analysis was performed following all applicable procedural requirements as per SOP/DP/QC/045 by selecting the latest calibration curve 'CAL_21032026.2026_03_21_11_54_09.cald'

Sr. No.	Name of Sample	No. of Sample to be injected
01	Same Vial	01
02	Refilled Vial	01
03	Re-Diluted Vial	01

After completion of above run, verified the obtained results and noted that SST results obtained from same vial found comparable with the initially obtained results and results obtained from 'Refilled Vial' found slightly lower than the initial results, however results obtained from 'Re-Diluted Vial' found at lower side significantly. Refer Attachment IX, X and XI.

Name of Vials	Hypothesis Testing - System Suitability Test Results			
	Recovery of 1,4-Benzoquinone	Mean Area	SD (NMT 0.5)	CV (NMT 5.0 %)
Same Vial	117.0 %	48.91	0.6486	1.33 %
Refilled Vial	104.6 %	44.27	0.9340	2.11 %
Re-Diluted Vial	76.26 %	33.76	0.3251	0.96 %
(SST Limit: Between 85-115 %)				

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Based on the outcome of recommended action – I, it is observed that system suitability results obtained from ‘Same vial’ does not meet the acceptance (Limit: 85 % to 115 %). However, results obtained from ‘Rcfill vial’ meet the acceptance and there was variability in obtained results as SST results obtained from ‘Re-diluted vial’ found very less as compare with the initial SST results and it might be happened probably due to uneven mixing or improper preparation of the standard solution. Hence to rule out the preparation error, additional investigation followed by hypothesis testing shall be carried out.

Recommended action II:

Based on the probable cause, it is recommended to prepare the fresh solution of 1,4 – Benzoquinone by taking the new vial of 1,4 –Benzoquinone USPRS, Lot No: R15690 (In house Vial No.: RS/002/23-003) as below under the supervision of QC reviewer.

Preparation of Benzoquinone stock standard solution (50000 ppb):

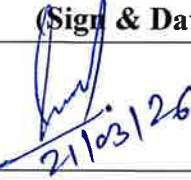
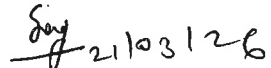
Weigh 7.5 mg of 1,4 -Benzoquinone & transfer in 100 ml volumetric flask, then add 50 ml of purified water in above 100 ml volumetric flask, sonicate for 30 seconds to dissolve completely and then make up the final volume to 100 ml.

Preparation of 1,4 -Benzoquinone Working Standard (500 ppb):

Take 1 ml of above stock standard (50000 ppb) by pipetting with Micropipette/ Glass pipette in 100 ml in a volumetric flask and make up the final volume to 100 ml with TOC water.

Inject the freshly prepared 1,4 -Benzoquinone (500 ppb) solution as per below sequence.

Sr. No.	Name of Sample	No. of Sample to be injected
01	1,4 -Benzoquinone	01

Prepared By (User) (Sign & Date)	Reviewed By (User) (Sign & Date)	Reviewed By (QA) (Sign & Date)
 21/03/26	 21/03/26	 21/03/26

21-03-2026 19:46:07(+05:30) : Confirm[Dhananjay Ibhad]Other CommentData Submitted For Review

21-03-2026 21:49:24(+05:30) : Review[Nilesh Dushing]Other CommentData Reviewed

Date of creation 2026/03/21 19:41:52
Sample Run TOC_I-QC-018_2026 - 5-570-4 - CAL_21032026_02.txl
User Dhananjay Ibhad
Object ID 0L-H54436004105-13525E12EECD-0000
System TOC-L + ASI-L
Origin TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald

Sample

Sample Name: 1,4-Benzoquinone
Sample ID: Std SST
Object ID 0L-H54436004105-13525E12EECD-0000
Origin: TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald
Status Completed
Chk. Result 79.76% : USP/EP system suitability test: Fail
Comment

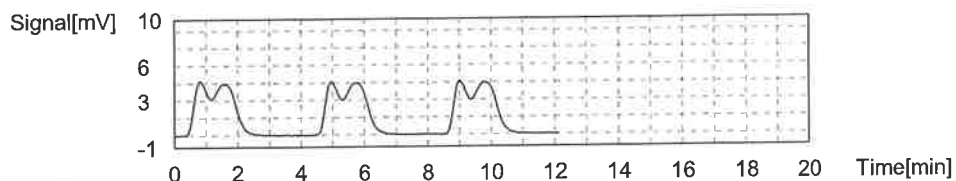
Type	Anal	Manual Dilution	Result
Unknown	NPOC	1.000	NPOC:471.9ppb

1. Det

Anal.: NPOC

No	Area	Conc.	Inj. Vol.	Aut. Dil.	Rem.	Ex.	Cal Curve	Date / Time
1	34.95	470.4ppb	816uL	1.000	*****		TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald	2026/03/21 19:33:01
2	35.25	474.5ppb	816uL	1.000	*****		TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald	2026/03/21 19:37:29
3	34.98	470.8ppb	816uL	1.000	*****		TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald	2026/03/21 19:41:51

Acid Add. 0.000%
Spurge Gas Flow 80mL/min
Sp. Time 60.00sec
Mean Area 35.06
Mean Conc. 471.9ppb
SD Area 0.1652
CV Area 0.47%
SD Conc 2.224
CV Conc 0.47%
Vial 6
No Washes 2

**History**

12-03-2026 18:19:19(+05:30) : Confirm[Dhananjay lbhad]Other CommentData Submitted For Review

12-03-2026 18:29:07(+05:30) : Review[pravin kolkand]Other CommentData Reviewed

Date of creation	2026/03/12 16:20:03
Sample Run	TOC_I-QC-018_2026 - 5-553-4 - CAL_12032026_03.tlx
User	Dhananjay lbhad
Object ID	0L-H54436004105-13525D825B8F-0000
System	TOC-L + ASI-L
Origin	TOC_I-QC-018_2026 - CAL_12032026_01.2026_03_12_15_48_20.cald

Cal. Curve

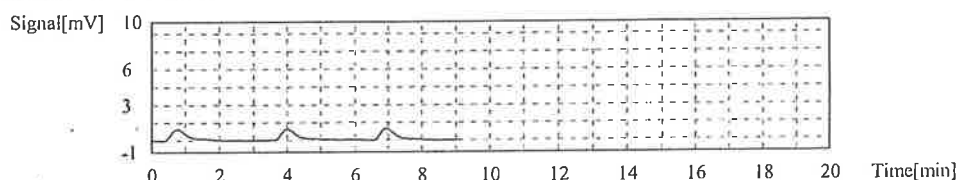
Sample Name:	Sucrose
Sample ID:	Standard
Object ID	0L-H54436004105-13525D825B8F-0000
Cal. Curve:	TOC_I-QC-018_2026 - CAL_12032026_01.2026_03_12_15_48_20.cald
Status	Completed
Comment:	

Type	Anal
Standard	NPOC

Conc: 0.000ppb

No.	Area	Inj. Vol.	Aut. Dil.	Rem.	Ex.	Date / Time
1	3.468	816uL	1.000	*****		2026/03/12 15:56:31
2	3.006	816uL	1.000	*****		2026/03/12 15:59:51
3	2.986	816uL	1.000	*****		2026/03/12 16:03:09

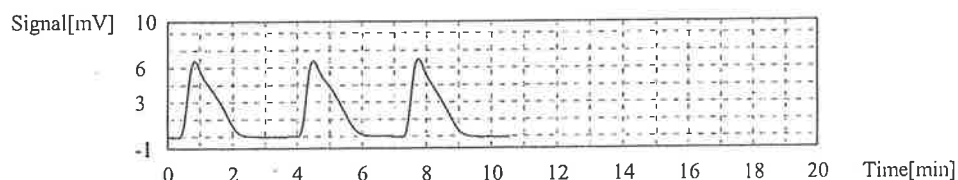
Acid Add. 0.000%
 Spurge Gas Flow 80mL/min
 Sp. Time 60.00sec
 Mean Area 3.153
 SD Area 0.2727
 CV Area 8.65%
 Vial 1
 No Washes 2



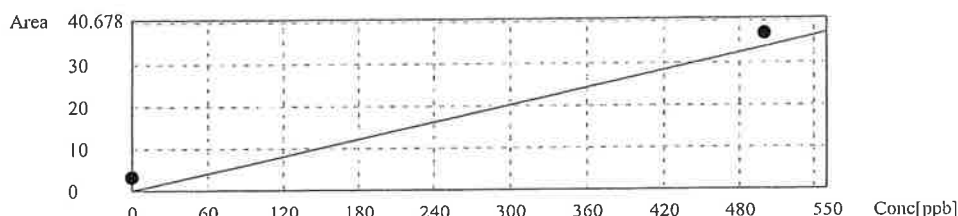
Conc: 500.0ppb

No.	Area	Inj. Vol.	Aut. Dil.	Rem.	Ex.	Date / Time
1	36.68	816uL	1.000	*****		2026/03/12 16:12:22
2	37.26	816uL	1.000	*****		2026/03/12 16:16:01
3	37.00	816uL	1.000	*****		2026/03/12 16:20:02

Acid Add. 0.000%
 Spurge Gas Flow 80mL/min
 Sp. Time 60.00sec
 Mean Area 36.98
 SD Area 0.2905
 CV Area 0.79%
 Vial 2
 No Washes 2



Slope: N/A
 Intercept N/A
 r² N/A
 r N/A
 RSE(%)
 Zero Shift Yes



History

No.	1
Date & Time	2026/03/12 15:33:05(+0530)
User Name	Dhananjay lbhad
Changes	New
Contents	
Reason for change	

12-03-2026 18:19:19(+05:30) : Confirm[Dhananjay Ibhad]Other CommentData Submitted For Review

12-03-2026 18:29:06(+05:30) : Review[pravin kolkand]Other CommentData Reviewed

Date of creation	2026/03/12 16:44:04
Sample Run	TOC_I-QC-018_2026 - 5-553-7 - CAL_12032026_03.tlx
User	Dhananjay Ibhad
Object ID	0L-H54436004105-13525D827AC9-0000
System	TOC-L + ASI-L
Origin	TOC_I-QC-018_2026 - CAL_12032026_01.2026_03_12_15_48_20.cald

Sample

Sample Name:	1,4 Benzoquinone
Sample ID:	Std SST
Object ID	0L-H54436004105-13525D827AC9-0000
Origin:	TOC_I-QC-018_2026 - CAL_12032026_01.2026_03_12_15_48_20.cald
Status	Completed
Chk. Result	98.44% : USP/EP system suitability test: Pass
Comment	

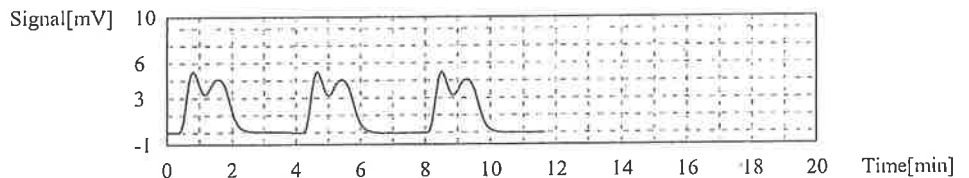
Type	Anal.	Manual Dilution	Result
Unknown	NPOC	1.000	NPOC:538.8ppb

1. Det

Anal.: NPOC

No.	Area	Conc.	Inj Vol.	Aut. Dil.	Rem.	Ex.	Cal Curve	Date / Time
1	36.03	532.6ppb	816uL	1.000	**h****		TOC_I-QC-018_2026 - CAL_12032026_01.2026_03_12_15_48_20.cald	2026/03/12 16:35:29
2	36.81	544.1ppb	816uL	1.000	**h****		TOC_I-QC-018_2026 - CAL_12032026_01.2026_03_12_15_48_20.cald	2026/03/12 16:39:44
3	36.52	539.8ppb	816uL	1.000	**h****		TOC_I-QC-018_2026 - CAL_12032026_01.2026_03_12_15_48_20.cald	2026/03/12 16:44:03

Acid Add.	0.000%
Spurge Gas Flow	80mL/min
Sp. Time	60.00sec
Mean Area	36.45
Mean Conc.	538.8ppb
SD Area	0.3943
CV Area	1.08%
SD Conc	5.828
CV Conc	1.08%
Vial	3
No Washes	2



History